UNSW Mathematics Society × Competitive Programming and Mathematics Society Presents

MATH3611/5705 Seminar

Presented by Jeff, Brendan and Daniel

Attendance

Please scan the QR code; please let us know about any dietary requirements (for the pizza!).

1. Sets and Cardinality

Cardinality

Comparing Sizes of Sets

Suppose that A and B are sets:

(i) If there exists an injective map $f : A \to B$, then $|A| \leq |B|$.

(ii) If there exists a surjective map $f : A \to B$, then $|A| \geq |B|$.

(iii) If there exists a bijective map $f : A \to B$, then $|A| = |B|$.

(iv) If $|A| \leq |B|$ and $|A| \neq |B|$, then $|A| < |B|$.

Schröder-Bernstein Theorem

If $|A| \leq |B|$ and $|B| \leq |A|$ then $|A| = |B|$.

Axiom of Choice

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Suppose that $\{S_i\}_{i \in I}$ is a non-empty collection of non-empty sets.

Then the Cartesian product

$$\prod_{i \in I} S_i$$

is also non-empty.

Countability

Countable vs Uncountable

A set S is:

- Countable if $|S| \leq |\mathbb{N}|$.
- Finite if $|S| < |\mathbb{N}|$.
- Countably infinite if $|S| = |\mathbb{N}|$.
- Uncountable if $|S| > |\mathbb{N}|$.

Dedekind-finiteness

A set S is said to be Dedekind-infinite if there is a bijection from S to a proper subset of itself. Otherwise, S is Dedekind-finite.

Equivalence of Dedekind-finiteness and Finiteness

A set is Dedekind-finite if and only if it is finite.

2019 Q1

Exam Question

Let S be the set of strictly increasing sequences of natural numbers. Determine whether S is countable or uncountable.

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Suppose that S is countable. Then there is a surjective function $f : \mathbb{N} \to S$. We have

$$\begin{aligned} f(0) &= (f_{0,0}, f_{0,1}, \dots) \\ f(1) &= (f_{1,0}, f_{1,1}, \dots) \\ &\vdots \end{aligned}$$

2019 Q1

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Suppose that S is countable. Then there is a surjective function $f : \mathbb{N} \to S$. We have

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f(0) &= (f_{0,0}, f_{0,1}, \dots) \\
f(1) &= (f_{1,0}, f_{1,1}, \dots) \\
&\vdots
\end{aligned}
$$

Consider $(f_0, f_1, \dots)$ where $f_0 = f_{0,0} + 1$ and $f_n = f_{n-1} + f_{n,n}$. This sequence of strictly increasing natural numbers is not in f since it is different to $f(n)$ since $f_{n,n} \neq f_n$. This contradicts our claim that f is surjective so S is uncountable.

Countability of Finite Products

Question

Suppose that $\{S_i\}_{i=1}^n$ is a family of countable sets. Then

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Since S_i is countable, we have an injective function $f_i : S_i \to \mathbb{N}$. We can make the injective function $f : \prod_{i=1}^n S_i \to \prod_{i=1}^n \mathbb{N}$ where

$$f(x_1, x_2, \ldots, x_n) = (f_1(x_1), f_2(x_2), \ldots, f_n(x_n)).$$

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We can simply show that $\prod_{i=1}^n \mathbb{N}$ is countable. This can be done through induction. The base case $n = 1$ is $\mathbb{N}$, which is countable.

Countability of Finite Products

Suppose that $\prod_{i=1}^{k} \mathbb{N}$ is countable. There is an injective function $g : \prod_{i=1}^{k} \mathbb{N} \to \mathbb{N}$. Consider the function $h : \prod_{i=1}^{k+1} \mathbb{N} \to \mathbb{N} \times \mathbb{N}$

$$h(x_1, x_2, \ldots, x_k, x_{k+1}) = (g(x_1, x_2, \ldots, x_k), x_{k+1}).$$

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$$h(x_1, x_2, \ldots, x_k, x_{k+1}) = (g(x_1, x_2, \ldots, x_k), x_{k+1}).$$

When $(g(x_1, x_2, \ldots, x_k), x_{k+1}) = (g(y_1, y_2, \ldots, y_k), y_{k+1})$, we can equate components to see $x_{k+1} = y_{k+1}$ and $g(x_1, x_2, \ldots, x_k) = g(y_1, y_2, \ldots, y_k)$. Since g is injective, we have $x_i = y_i$ for all $1 \le i \le k + 1$.

Countability of Finite Products

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The function h is injective so

$$\left| \prod_{i=1}^{k+1} \mathbb{N} \right| \leq |\mathbb{N} \times \mathbb{N}| = |\mathbb{N}|.$$

Tools for Countability

Countable Union of Countable Sets

Suppose that I is a nonempty countable set and $\{S_i\}_{i \in I}$ is a family of countable sets. Then

$$S = \bigcup_{i \in I} S_i$$

is countable.

Cantor's Theorem

For all sets S, $|S| < |\mathcal{P}(S)|$.

2014 Q1b

Question

Is the set $S = \{z \in \mathbb{C} : z^2 + mz + n^3 = 0 \text{ for some } m, n \in \mathbb{Z}\}$ countable or uncountable?

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For all $z \in S$, we have

$$z = \frac{-m + (-1)^k \sqrt{m^2 - 4n^3}}{2}$$

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For all $z \in S$, we have

$$z = \frac{-m + (-1)^k \sqrt{m^2 - 4n^3}}{2}$$

for some integers k, m, n. This means that we have a surjective function $f : \mathbb{Z} \times \mathbb{Z} \times \mathbb{Z} \to S$

$$f(k, m, n) = \frac{-m + (-1)^k \sqrt{m^2 - 4n^3}}{2}.$$

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Since $\mathbb{Z}$ is countable, we find that $\mathbb{Z} \times \mathbb{Z} \times \mathbb{Z}$ is countable. We have

$$|S| \leq |\mathbb{Z} \times \mathbb{Z} \times \mathbb{Z}| \leq |N|.$$

2. Metric Spaces

Metric Spaces

Metric Space Definition

Let X be a non-empty set. A function $d : X \times X \to [0, \infty)$ is a metric if it satisfy the following conditions.

(1) $d(x, y) = 0 \iff x = y$.

(2) (Symmetry) $d(x, y) = d(y, x)$ for all $x, y \in X$

(3) (Triangle inequality) $d(x, y) \leq d(x, z) + d(z, y)$ for all $x, y, z \in X$.

The pair (X, d) is called a metric space.

Go-To Examples of Metric Spaces

The following metrics should be in your back pocket.

1. The p-metric on $\mathbb{R}^n$ for $p \geq 1$
$$d(\mathbf{x}, \mathbf{y}) = \left(\sum_{k=1}^{n} |x_k - y_k|^p \right)^{1/p}.$$
Similar metrics can also be defined for functions and sequences (on spaces where they exist).
2. The discrete metric
$$d(\mathbf{x}, \mathbf{y}) = \begin{cases} 0 & \text{if } \mathbf{x} = \mathbf{y}, \\ 1 & \text{otherwise.} \end{cases}$$
3. The d_∞ metric on $\mathbb{R}^n$, where
$$d_\infty(\mathbf{x}, \mathbf{y}) = \sup_{1 \leq k \leq n} |x_k - y_k|.$$

Some Inequalities

Analysis is built on inequalities. In particular, the triangle inequality for the p-metric is known as *Minkowski's inequality*.

Minkowski's Inequality

Suppose that $p \geq 1$. Then, on $X = \mathbb{R}^n$, for any $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbb{R}^n$, we have

$$d_p(\mathbf{x}, \mathbf{y}) \leq d_p(\mathbf{x}, \mathbf{z}) + d_p(\mathbf{z}, \mathbf{y}).$$

This is typically proved using Hölder's inequality.

Hölder's Inequality

Suppose that $p \geq 1$ and q satisfies $\frac{1}{p} + \frac{1}{q} = 1$. Then, for any $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$,

$$\left| \sum_{i=1}^{n} x_i y_i \right| \leq \left(\sum_{i=1}^{n} |x_i|^p \right)^{1/p} \left(\sum_{i=1}^{n} |y_i|^q \right)^{1/q}$$

2017 Exam Question 3

Exam Questions

Suppose that (X, d) is a metric space. Define the function $\delta : X \times X \to \mathbb{R}$ by

$$\delta(x, y) = \min(d(x, y), 1) = \begin{cases} d(x, y), & d(x, y) < 1, \\ 1, & d(x, y) \geq 1. \end{cases}$$

(a) Prove that δ is a metric on X.

(b) Prove that a set $Y \subseteq X$ is open in (X, d) if and only if it is open in (X, δ).

2017 Exam Question 3

(a) Checking the conditions for a metric:

1. From the definition of δ and since $d(x,y) \geq 0$, $\delta(x,y) \geq 0$ for all $x, y \in X$. If $\delta(x,y) = 0$, then $\delta(x,y) = d(x,y)$, so $d(x,y) = 0$, and hence $x = y$.
2. Symmetry follows since $d(x,y) = d(y,x)$ for all $x, y \in X$, so

$$\delta(x,y) = \min(d(x,y), 1) = \min(d(y,x), 1) = \delta(y,x).$$

3. For the triangle inequality, let $x, y, z \in X$. If $\delta(x,y)$ and $\delta(y,z)$ are both less than 1, then

$$\delta(x,z) \leq d(x,z) \leq d(x,y) + d(y,z) = \delta(x,y) + \delta(y,z).$$

Otherwise,

$$\delta(x,z) \leq 1 \leq \delta(x,y) + \delta(y,z).$$

2017 Exam Question 3

(b) Denote the balls in the two metrics by $B_d(x,\varepsilon)$ and $B_\delta(x,\varepsilon)$. Since $\delta(x,y) \leq d(x,y)$ for all $x, y \in X$,$B_d(x,\varepsilon) \subseteq B_\delta(x,\varepsilon)$ for all $x \in X$ and $\varepsilon > 0$. Also,$B_\delta(x,\varepsilon) = B_d(x,\varepsilon)$ for $0 < \varepsilon < 1$.

- Suppose first that Y is open in (X,d) and $x \in Y$. Then there exists $\varepsilon > 0$ such that $B_d(x,\varepsilon) \subseteq Y$. Choose $0 < \varepsilon' < \min(1,\varepsilon)$, so that $B_d(x,\varepsilon') \subseteq Y$, and hence

$$B_\delta(x,\varepsilon') = B_d(x,\varepsilon') \subseteq Y.$$

  Thus x is an interior point of Y in (X,δ). Hence Y is open in (X,δ).
- Conversely, suppose Y is open in (X,δ) and $x \in Y$. Then there exists $\varepsilon > 0$ such that $B_\delta(x,\varepsilon) \subseteq Y$. Since $B_d(x,\varepsilon) \subseteq B_\delta(x,\varepsilon)$, we have $B_d(x,\varepsilon) \subseteq Y$. Thus x is an interior point of Y in (X,d), so Y is open in (X,d).

Convergence

Convergence Definition

Let (X, d) be a metric space. Suppose $\{x_k\}_{k=1}^{\infty} \subseteq X$ and that $x \in X$.

Then $\{x_k\}_{k=1}^{\infty}$ converges to x if

$$d(x_k, x) \to 0 \text{ as } k \to \infty.$$

As an $\epsilon - \delta$ definition,

$$\forall \epsilon > 0,\ \exists K \text{ such that if } k \geq K \text{ then } d(x_k, x) < \epsilon.$$

Note that limits in a metric space are unique.

Classifying Points

The $\epsilon - ball$ centred at x_0 is $B(x_0, \epsilon) = \{x \in X : d(x_0, x) < \epsilon\}$.

Types of Points

Let (X, d) be a metric space. Suppose that $Y \subseteq X$ and $x_0 \in Y$.

(i) x_0 is an *interior* point of Y if $B(x_0, \varepsilon) \subseteq Y$ for some $\epsilon > 0$.

(ii) x_0 is a *boundary* point of Y if for all $\varepsilon > 0$,

$$B(x_0, \varepsilon) \cap Y \neq \varnothing \text{ and } B(x_0, \varepsilon) \cap X \backslash Y \neq \varnothing.$$

(iii) x_0 is a *limit* point of Y if every $B(x_0, \varepsilon)$ contains some point of Y, different from x_0.

Boundedness

Boundedness

Under the following equivalent conditions for a metric space (X, d), we say that a set $\varnothing \neq Y \subseteq X$ is *bounded*:

1. For every $x \in X$, there exists $R(x) > 0$ such that
$$Y \subseteq B(x, R(x)).$$
2. There exist $y \in Y$ and $R > 0$ such that
$$Y \subseteq B(y, R).$$
3. There exists $R > 0$ such that for all $y_1, y_2 \in Y$,
$$d(y_1, y_2) < R.$$

Classifying Sets

Let (X, d) be a metric space. Suppose that $Y \subseteq X$.

Open Sets

The following are equivalent:

- Y is open.
- Y only contains its interior points.
- Y contains none of its boundary points.

Closed Sets

The following are equivalent:

- Y is closed.
- $X \backslash Y$ is open.
- Y contains all of its boundary points.
- Y contains all of its limit points.

Interior, Boundary and Closure

Interior, Boundary and Closure

For $Y \subseteq X$ in a metric space (X, d), denote

- the interior $\mathrm{Int}(Y)$ as the set of interior points of Y;
- the boundary $\mathrm{Bd}(Y)$ as the set of boundary points of Y;
- the closure $\mathrm{Cl}(Y)$ as the union of Y with its boundary.

Result on Boundary and Interior

Denoting $\sqcup$ as disjoint union, for any set $Y \subseteq X$ in a metric space (X, d),

$$X = \mathrm{Int}(Y) \sqcup \mathrm{Bd}(Y) \sqcup \mathrm{Int}(X \setminus Y).$$

Further,

- The set Y is open if and only if $Y = \mathrm{Int}(Y)$.
- The set Y is closed if and only if $Y = \mathrm{Int}(Y) \sqcup \mathrm{Bd}(Y) = \mathrm{Cl}(Y)$.

Continuity

Continuity Definition

Let (X, d_X) and (Y, d_Y) be metric spaces and suppose that $f : X \to Y$.

(1) f is continuous at $x_0 \in X$ if for all $\varepsilon > 0$, there exist $\delta > 0$ such that if $d_X(x, x_0) < \delta$ then $d_Y(f(x), f(x_0)) < \varepsilon$.

(2) f is continuous on X if f is continuous at every point of X.

Open Set Continuity

Let (X, d_X) and (Y, d_Y) be metric spaces and suppose that $f : X \to Y$. Then f is continuous on X iff $f^{-1}(U)$ is an open set in (X, d_X) for every open set U in (Y, d_Y).

Normed Spaces

Norm Definition

Let V be a vector space over $\mathbb{F} = \mathbb{R}$ or $\mathbb{F} = \mathbb{C}$. A function $\|\cdot\| : V \to \mathbb{R}$ is a norm if

(i) $\|v\| \geq 0$ for all $v \in V$, with $\|v\| = 0$ iff $v = 0$.

(ii) $\|\lambda v\| = |\lambda|\|v\|$ for all $\lambda \in \mathbb{F}$ and $v \in V$.

(iii) $\|u + v\| \leq \|u\| + \|v\|$ for all $u, v \in V$.

Setting $d(u, v) = \|u - v\|$ gives a metric on V.

2015 Q2

Question

Let X be $\mathbb{R}^2$ with the usual Euclidean norm topology, is the following closed or open?

$$Y = \bigcup_{k=1}^{\infty} \left\{ \boldsymbol{x} \in X : \|\boldsymbol{x} - (1/k, 0)\| \leq \frac{1}{k+1} \right\}$$

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This is a union of closed balls centered at $(1/k, 0)$ with radius $\frac{1}{k+1}$.

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This is a union of closed balls centered at $(1/k, 0)$ with radius $\frac{1}{k+1}$. These balls always overlap since the distance between centres is

$$\frac{1}{k} - \frac{1}{k+1} = \frac{1}{k(k+1)} \leq \frac{1}{k+1}.$$

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$$\frac{1}{k} - \frac{1}{k+1} = \frac{1}{k(k+1)} \leq \frac{1}{k+1}.$$

We see Y is not open since it contains a boundary point $(\frac{3}{2}, 0)$. This is because $B(\frac{3}{2}, \epsilon)$ contains $(\frac{3}{2} - \frac{\epsilon}{2}, 0) \in Y$ and $(\frac{3}{2} + \frac{\epsilon}{2}, 0) \in X \backslash Y$.

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We see Y is not open since it contains a boundary point $(\frac{3}{2}, 0)$. This is because $B(\frac{3}{2}, \epsilon)$ contains $(\frac{3}{2} - \frac{\epsilon}{2}, 0) \in Y$ and $(\frac{3}{2} + \frac{\epsilon}{2}, 0) \in X \backslash Y$.

Each closed ball does not contain $(0, 0)$. Since Y contains all points of the form $\frac{1}{k}$, we can get arbitrarily close to Y so it is a limit point.

Inner Product Spaces

Inner Product Definition

An inner product on a vector space V over $\mathbb{F} = \mathbb{C}$ or $\mathbb{F} = \mathbb{R}$ is a map $\langle \cdot, \cdot \rangle : V \times V \to \mathbb{F}$ such that

(i) $\langle v, v \rangle \geq 0$ for all $v \in V$, with $\langle v, v \rangle = 0$ iff $v = 0$.

(ii) $\langle \lambda v, u \rangle = \lambda \langle v, u \rangle$ for all $\lambda \in \mathbb{F}$ and $u, v \in V$.

(iii) $\langle u, v \rangle = \overline{\langle v, u \rangle}$ for all $u, v \in V$.

(iv) $\langle u + v, w \rangle = \langle u, w \rangle + \langle v, w \rangle$ for all $u, v, w \in V$.

Setting $\|v\| = \sqrt{\langle v, v \rangle}$ gives a norm on V.

A vector space V equipped with an inner product is called an **inner product space**, denoted by $(V, \langle \cdot, \cdot \rangle)$.

Completeness

Cauchy Sequences

Let $\{x_k\}_{k=1}^{\infty}$ be a sequence in a metric space (X, d). We say that $\{x_n\}$ is a Cauchy sequence if for all $\varepsilon > 0$ there exists K such that for all $k, l \geq K$, $d(x_k, x_l) < \varepsilon$.

Completeness

Cauchy Sequences

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Completeness

A metric space (X, d) is said to be complete if every Cauchy sequence in X converges to an element in X.

Types of Metric Spaces

1. A Banach space is a complete normed vector space.
2. A Hilbert space is a complete inner product space.

The ℓ^p Space

The ℓ^p Space

Define the following set:

$$\ell^p = \left\{ \{x_k\}_{k=1}^\infty \subseteq \mathbb{R} : \sum_{k=1}^\infty |x_k|^p < \infty \right\}.$$

Completeness of ℓ^p

This is a Banach space with norm

$$d_p(\{x_k\}_{k=1}^\infty) = \left(\sum_{k=1}^\infty |x_k|^p \right)^{\frac{1}{p}}.$$

(The vector space operations are defined pointwise.)

The ℓ^∞ Space

The ℓ^∞ Space

Define the following set:

$$\ell^\infty = \left\{ \{x_k\}_{k=1}^\infty \subseteq \mathbb{R} : \sup_k |x_k| < \infty \right\}.$$

Completeness of ℓ^∞

This is a Banach space with norm

$$d_\infty(\{x_k\}_{k=1}^\infty) = \sup_k |x_k|.$$

(The vector space operations are defined pointwise.)

The c_0 and c_{00} spaces

Important subsets of ℓ^∞

Define the following two important subsets of ℓ^∞:

1. The set c_0 of sequences $\{x_k\}_{k=1}^\infty$ with $\lim\limits_{k\to\infty} x_k = 0$. (This is contained in ℓ^∞, but not in every ℓ^p.)
2. The set c_{00}, the vector space of real sequences $\{x_n\}_{n=1}^\infty$ such that $x_n = 0$ for all but finitely many n. This is contained in every ℓ^p and in c_0.

Properties of c_0 and c_{00}

The following statements are true:

(a) The space c_{00} is dense in ℓ^p under the $\|\cdot\|_p$ norm.

(b) The closure of c_{00} is c_0 under the ℓ^∞ metric.

Cauchy Test

Recall that part of what makes the definition of completeness important is the following:

Cauchy Test

Every convergent sequence $\{x_k\}_{k=1}^{\infty}$ in a metric space (X, d) is Cauchy, that is,

$$\lim_{n,m \to \infty} d(x_m, x_n) = 0.$$

This means that in a complete space, being Cauchy is a necessary and sufficient criterion for convergence.

2018 Q10

Exam Question

Let $(H, \langle \cdot, \cdot \rangle)$ be a Hilbert space. Suppose that $\{\boldsymbol{x}_k\}_{k=1}^{\infty}$ is an orthonormal set in H and that $\boldsymbol{y} \in H$. For $n = 1, 2, 3, \ldots$, let

$$\boldsymbol{y}_n = \sum_{k=1}^{n} \langle \boldsymbol{x}_k, \boldsymbol{y} \rangle \boldsymbol{x}_k \text{ and } \boldsymbol{z}_n = \boldsymbol{y} - \boldsymbol{y}_n.$$

(a) Prove that the sequence $(\boldsymbol{x}_k)_{k=1}^{\infty}$ is not Cauchy in H.

(b) Prove that for each n, $\|\boldsymbol{y}\|^2 = \|\boldsymbol{y}_n\|^2 + \|\boldsymbol{z}_n\|^2$.

2018 Q10

Exam Question

Let $(H, \langle \cdot, \cdot \rangle)$ be a Hilbert space. Suppose that $\{\boldsymbol{x}_k\}_{k=1}^{\infty}$ is an orthonormal set in H.

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(a) Prove that the sequence $(\boldsymbol{x}_k)_{k=1}^{\infty}$ is not Cauchy in H.

Consider for integers k, l,

$$\begin{aligned}
\|\boldsymbol{x}_k - \boldsymbol{x}_l\|^2 &= \langle \boldsymbol{x}_k - \boldsymbol{x}_l, \boldsymbol{x}_k - \boldsymbol{x}_l \rangle \\
&= \langle \boldsymbol{x}_k, \boldsymbol{x}_k \rangle - 2\langle \boldsymbol{x}_k, \boldsymbol{x}_l \rangle + \langle \boldsymbol{x}_l, \boldsymbol{x}_l \rangle \\
&= \|\boldsymbol{x}_k\|^2 - 0 + \|\boldsymbol{x}_l\|^2 = 2,
\end{aligned}$$

since orthonormal means $\langle \boldsymbol{x}_k, \boldsymbol{x}_l \rangle = 0$ and each norm is 1.

2018 Q10

Exam Question

Let $(H, \langle \cdot, \cdot \rangle)$ be a Hilbert space. Suppose that $\{\boldsymbol{x}_k\}_{k=1}^{\infty}$ is an orthonormal set in H.

(a) Prove that the sequence $(\boldsymbol{x}_k)_{k=1}^{\infty}$ is not Cauchy in H.

Consider for integers k, l,

$$\begin{aligned} \|\boldsymbol{x}_k - \boldsymbol{x}_l\|^2 &= \langle \boldsymbol{x}_k - \boldsymbol{x}_l, \boldsymbol{x}_k - \boldsymbol{x}_l \rangle \\ &= \langle \boldsymbol{x}_k, \boldsymbol{x}_k \rangle - 2\langle \boldsymbol{x}_k, \boldsymbol{x}_l \rangle + \langle \boldsymbol{x}_l, \boldsymbol{x}_l \rangle \\ &= \|\boldsymbol{x}_k\|^2 - 0 + \|\boldsymbol{x}_l\|^2 = 2, \end{aligned}$$

since orthonormal means $\langle \boldsymbol{x}_k, \boldsymbol{x}_l \rangle = 0$ and each norm is 1.

We see $\|\boldsymbol{x}_k - \boldsymbol{x}_l\|^2 = 2$ which does not get arbitrarily small so the sequence is not Cauchy.

2018 Q10

Exam Question

Let $(H, \langle\cdot,\cdot\rangle)$ be a Hilbert space. Suppose that $\{\boldsymbol{x}_k\}_{k=1}^{\infty}$ is an orthonormal set in H and that $\boldsymbol{y} \in H$. For $n = 1, 2, 3, \ldots$, let

$$\boldsymbol{y}_n = \sum_{k=1}^{n} \langle \boldsymbol{x}_k, \boldsymbol{y} \rangle \boldsymbol{x}_k \text{ and } \boldsymbol{z}_n = \boldsymbol{y} - \boldsymbol{y}_n.$$

(b) Prove that for each n, $\|\boldsymbol{y}\|^2 = \|\boldsymbol{y}_n\|^2 + \|\boldsymbol{z}_n\|^2$.

2018 Q10

Exam Question

Let $(H, \langle \cdot, \cdot \rangle)$ be a Hilbert space. Suppose that $\{\boldsymbol{x}_k\}_{k=1}^{\infty}$ is an orthonormal set in H and that $\boldsymbol{y} \in H$. For $n = 1, 2, 3, \ldots$, let

$$\boldsymbol{y}_n = \sum_{k=1}^{n} \langle \boldsymbol{x}_k, \boldsymbol{y} \rangle \boldsymbol{x}_k \text{ and } \boldsymbol{z}_n = \boldsymbol{y} - \boldsymbol{y}_n.$$

(b) Prove that for each n, $\|\boldsymbol{y}\|^2 = \|\boldsymbol{y}_n\|^2 + \|\boldsymbol{z}_n\|^2$.

We have $\boldsymbol{y} = \boldsymbol{y}_n + \boldsymbol{z}_n$ so $\|\boldsymbol{y}\|^2 = \|\boldsymbol{y}_n\|^2 + 2\langle \boldsymbol{y}_n, \boldsymbol{z}_n \rangle + \|\boldsymbol{y}_n\|^2$. We see

$$\begin{aligned}
\langle \boldsymbol{y}_n, \boldsymbol{z}_n \rangle &= \left\langle \sum_{k=1}^{n} \langle \boldsymbol{x}_k, \boldsymbol{y} \rangle \boldsymbol{x}_k, \boldsymbol{y} - \sum_{j=1}^{n} \langle \boldsymbol{x}_j, \boldsymbol{y} \rangle \boldsymbol{x}_j \right\rangle \\
&= \sum_{k=1}^{n} \left\langle \boldsymbol{x}_k, \boldsymbol{y} \rangle \langle \boldsymbol{x}_k, \boldsymbol{y} - \sum_{j=1}^{n} \langle \boldsymbol{x}_j, \boldsymbol{y} \rangle \boldsymbol{x}_j \right\rangle
\end{aligned}$$

2018 Q10

$$
\begin{aligned}
\langle \boldsymbol{y}_n, \boldsymbol{z}_n \rangle &= \sum_{k=1}^{n} \langle \boldsymbol{x}_k, \boldsymbol{y} \rangle \langle \boldsymbol{x}_k, \boldsymbol{y} \rangle - \sum_{k=1}^{n} \left\langle \boldsymbol{x}_k, \boldsymbol{y} \rangle \langle \boldsymbol{x}_k, \sum_{j=1}^{n} \langle \boldsymbol{x}_j, \boldsymbol{y} \rangle \boldsymbol{x}_j \right\rangle \\
&= \sum_{k=1}^{n} \langle \boldsymbol{x}_k, \boldsymbol{y} \rangle \langle \boldsymbol{x}_k, \boldsymbol{y} \rangle - \sum_{k=1}^{n} \sum_{j=1}^{n} \langle \boldsymbol{x}_j, \boldsymbol{y} \rangle \langle \boldsymbol{x}_k, \boldsymbol{y} \rangle \langle \boldsymbol{x}_k, \boldsymbol{x}_j \rangle \\
&= \sum_{k=1}^{n} \langle \boldsymbol{x}_k, \boldsymbol{y} \rangle \langle \boldsymbol{x}_k, \boldsymbol{y} \rangle - \sum_{k=1}^{n} \langle \boldsymbol{x}_k, \boldsymbol{y} \rangle \langle \boldsymbol{x}_k, \boldsymbol{y} \rangle \\
&= 0,
\end{aligned}
$$

since $\langle \boldsymbol{x}_k, \boldsymbol{x}_j \rangle = 0$ when $k \neq j$ and $\langle \boldsymbol{x}_k, \boldsymbol{x}_j \rangle = 1$ when $k = j$.

2018 Q9b

Exam Question

Is the set $Y = \{f \in C[0,1] : \int_0^1 f(t)\,dt \geq 5\}$ a complete metric space with the uniform metric d_∞?

2018 Q9b

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Complete Metric Space Subset

Let (X, d) be a complete metric space. Suppose that $Y \subseteq X$. Then (Y, d) is complete if and only if Y is closed in X.

2018 Q9b

Exam Question

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Complete Metric Space Subset

Let (X, d) be a complete metric space. Suppose that $Y \subseteq X$. Then (Y, d) is complete if and only if Y is closed in X.

We claim the interior of Y is $f \in C[0,1]$ where $\int_0^1 f(t)\,dx > 5$. Let $\epsilon = \int_0^1 f(t)\,dx - 5$ and consider $g \in B(f, \epsilon)$.

Exam Question

Is the set $Y = \{f \in C[0,1] : \int_0^1 f(t)\,dt \geq 5\}$ a complete metric space with the uniform metric d_∞?

Complete Metric Space Subset

Let (X, d) be a complete metric space. Suppose that $Y \subseteq X$. Then (Y, d) is complete if and only if Y is closed in X.

We claim the interior of Y is $f \in C[0,1]$ where $\int_0^1 f(t)\,dx > 5$. Let $\epsilon = \int_0^1 f(t)\,dx - 5$ and consider $g \in B(f, \epsilon)$. Consider that

$$\int_0^1 g(t)\,dt > \int_0^1 f(t) - \epsilon\,dt = \int_0^1 f(t)\,dt - \epsilon = 5,$$

since $d_\infty(g, f) < \epsilon$. Therefore $B(f, \epsilon)$ remains in the interior.

2018 Q9b

The interior of $X \backslash Y$ can be proven to be $\int_0^1 f(t)\,dx < 5$ with a similar method.

2018 Q9b

The interior of $X \backslash Y$ can be proven to be $\int_0^1 f(t)\,dx < 5$ with a similar method. The boundary of Y is $f \in C[0,1]$ where $\int_0^1 f(t)\,dx = 5$. For all $\epsilon > 0$, we have $f - \frac{\epsilon}{2}$ where

$$\int_0^1 f(t) - \frac{\epsilon}{2}\,dt = 5 - \frac{\epsilon}{2} < 5 \text{ so } f(t) - \frac{\epsilon}{2} \in B(f,\epsilon) \cap X \backslash Y$$

2018 Q9b

The interior of $X \backslash Y$ can be proven to be $\int_0^1 f(t)\,dx < 5$ with a similar method. The boundary of Y is $f \in C[0,1]$ where $\int_0^1 f(t)\,dx = 5$. For all $\epsilon > 0$, we have $f - \frac{\epsilon}{2}$ where

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and we have $f \in B(f,\epsilon) \cap Y$. The set Y is closed since it contains all its boundary points.

2018 Q9b

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and we have $f \in B(f,\epsilon) \cap Y$. The set Y is closed since it contains all its boundary points.

Since Y is closed in the complete metric space $(C[0,1], d_\infty)$, (Y, d) is a complete metric space.

Contraction Mappings

Contraction Map

Let (X, d) be a metric space and suppose that $f : (X, d) \to (X, d)$. We say that f is a contraction if there exists a constant $c \in [0, 1)$ such that for all $x, y \in X$,

$$d(f(x), f(y)) \leq c \times d(x, y).$$

Contraction Mapping Theorem

Let (X, d) be a complete metric space. If $f : (X, d) \to (X, d)$ is a contraction then there is a unique point $x \in X$ such that $f(x) = x$.

2018 Q5c

Exam Question

Let X be the interval $[0, \frac{1}{2})$ with the standard metric inherited from $\mathbb{R}$. Is the function

$$f : X \to X,\ f(x) = x^2$$

a contraction?

2018 Q5c

Exam Question

Let X be the interval $[0, \frac{1}{2})$ with the standard metric inherited from $\mathbb{R}$. Is the function

$$f : X \to X,\ f(x) = x^2$$

a contraction?

Suppose that there is some $c < 1$ such that

$$|x^2 - y^2| \leq c|x - y| \text{ for all } x, y \in X.$$

2018 Q5c

Exam Question

Let X be the interval $[0, \frac{1}{2})$ with the standard metric inherited from $\mathbb{R}$. Is the function

$$f : X \to X,\ f(x) = x^2$$

a contraction?

Suppose that there is some $c < 1$ such that

$$|x^2 - y^2| \leq c|x - y| \text{ for all } x, y \in X.$$

If $x \neq y$, then $|x + y| \leq c < 1$. Let $x = \frac{3c+1}{8}$ and $y = \frac{c+3}{8}$. We have

$$|x + y| = \frac{c+1}{2} > c \implies |x^2 - y^2| > c|x - y|,$$

2018 Q5c

Exam Question

Let X be the interval $[0, \frac{1}{2})$ with the standard metric inherited from $\mathbb{R}$. Is the function

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a contraction?

Suppose that there is some $c < 1$ such that

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If $x \neq y$, then $|x + y| \leq c < 1$. Let $x = \frac{3c+1}{8}$ and $y = \frac{c+3}{8}$. We have

$$|x + y| = \frac{c+1}{2} > c \implies |x^2 - y^2| > c|x - y|,$$

which is a contradiction so there is no such c and f is not a contraction.

2022 Exam Question 4

Exam Question

Suppose that we want to solve the functional equation

$$2f(x) - f(x+1) = \sin x, \qquad x \in \mathbb{R}.$$

By writing this as a fixed point problem on a suitable metric space, show that there is at least one continuous function f which satisfies this equation.

Let $X = C_b(\mathbb{R})$, the set of continuous and bounded functions from $\mathbb{R}$ to $\mathbb{R}$, with the metric d_∞. Define $T : X \to X$ by

$$T(f)(x) = \frac{1}{2}\big(\sin x - f(x+1)\big).$$

Then $T(f) \in X$, and $T(f) = f$ if and only if f satisfies the functional equation.

2022 Exam Question 4

Now, we have

$$\begin{aligned} d_\infty(T(f), T(g)) &= \sup_{x\in\mathbb{R}} |T(f)(x) - T(g)(x)| \\ &= \frac{1}{2} \sup_{x\in\mathbb{R}} |g(x+1) - f(x+1)| \\ &= \frac{1}{2} d_\infty(f, g). \end{aligned}$$

Thus T is a contraction on X with $c = \frac{1}{2}$. Since X is complete, the Contraction Mapping Theorem gives a unique $f \in X$ such that $T(f) = f$. Hence there is a continuous bounded solution.

There may be other continuous solutions if the boundedness restriction is removed.

2024 Final Exam Question 4(b)

Exam Question

Let $X = (0,1) \times (0,1)$ with the metric

$$d_1((x_1, y_1), (x_2, y_2)) = |x_1 - x_2| + |y_1 - y_2|.$$

Define $T : X \to X$ by

$$T(x, y) = \left(\frac{x+y}{3}, \frac{2x}{3} \right).$$

Is T a contraction on (X, d_1)?

2024 Final Exam Question 4(b)

What motivates us is attempting to see what the proof would look like. Let $x_1 = (x_1, y_1)$, $x_2 = (x_2, y_2) \in X$. Then

$$\begin{aligned} d_1(T(x_1), T(x_2)) &= \left| \frac{(x_1 + y_1) - (x_2 + y_2)}{3} \right| + \left| \frac{x_1^2 - x_2^2}{3} \right| \\ &\leq \frac{1}{3} \left(|x_1 - x_2| + |y_1 - y_2| + |x_1 - x_2||x_1 + x_2| \right) \\ &\leq d_1(x_1, x_2). \end{aligned}$$

However, we need a strict bound with some $c < 1$.

2024 Final Exam Question 4(b)

Consider the points

$$x_1 = (1-\epsilon, 0), \qquad x_2 = \left(1 - \frac{\epsilon}{2}, 0\right) \in X,$$

where $\epsilon > 0$ is small. Then

$$d_1(x_1, x_2) = \frac{\epsilon}{2},$$

while

$$d_1(T(x_1), T(x_2)) = \left(1 - \frac{\epsilon}{2}\right) d_1(x_1, x_2).$$

Given any $c < 1$, we can choose ϵ sufficiently small so that

$$1 - \frac{\epsilon}{2} > c.$$

Thus T is not a contraction.

Picard-Lindelöf Theorem

Lipschitz Continuous in the Second Variable

If $X \subseteq \mathbb{R}^2$, $f : X \to \mathbb{R}$ is Lipschitz continuous if there is a $K > 0$ where

$$|f(x, y_1) - f(x, y_2)| \leq K|y_1 - y_2|, \forall (x, y_1), (x, y_2) \in X.$$

Lipschitz Continuous Lemma

If f is continuous on $[a, b]$, differentiable on (a, b) and there is a $K > 0$ such that $|f'| \leq K$ on (a, b), then f is Lipschitz continuous on $[a, b]$.

Picard-Lindelöf Theorem

Let g be continuous on a neighbourhood of $(a, b) \in \mathbb{R}^2$ and Lipschitz continuous. Then

$$y' = g(x, y),\ y(a) = b$$

has a unique solution for an interval around a.

2021 Q4

Exam Question

What does the Picard-Lindelöf Theorem say about

$$y' = e^{-xy},\ y(0) = 4?$$

2021 Q4

Exam Question

What does the Picard-Lindelöf Theorem say about

$$y' = e^{-xy},\ y(0) = 4?$$

The function $g(x, y) = e^{-xy}$ is continuous since it is elementary.

2021 Q4

Exam Question

What does the Picard-Lindelöf Theorem say about

$$y' = e^{-xy},\ y(0) = 4?$$

The function $g(x, y) = e^{-xy}$ is continuous since it is elementary. We can consider

$$g'_y(x, y) = -xe^{-xy},$$

which is also a continuous function.

2021 Q4

Exam Question

What does the Picard-Lindelöf Theorem say about

$$y' = e^{-xy},\ y(0) = 4?$$

The function $g(x, y) = e^{-xy}$ is continuous since it is elementary. We can consider

$$g'_y(x, y) = -xe^{-xy},$$

which is also a continuous function. We can choose a neighbourhood of 0, where $g'_y(x, y)$ has a maximum value, by min-max theorem.

Therefore g is Lipschitz continuous and there is a unique solution for a neighbourhood around 0, by Picard-Lindelöf Theorem.

2021 Q4

More precisely, consider the set

$$Y = \{y \in C([-\delta,\delta]) : y(x) \in [4-\varepsilon, 4+\varepsilon] \text{ for all } x \in [-\delta,\delta]\}\,,$$

for some δ we will choose later.

Observe that Y is a closed set in the complete space $C([-\delta,\delta])$ under the $\|\cdot\|_\infty$ norm.

If

$$y' = e^{-xy}, \qquad y(0) = 4,$$

then y is a fixed point of the map

$$T : C([-\delta,\delta]) \to C([-\delta,\delta]),$$

where

$$T(y)(x) = 4 + \int_0^x e^{-ty(t)}\,dt.$$

With out

$$g(x,y) = e^{-xy}.$$

We first work with

$$[-\delta,\delta] \subseteq [-1,1], \qquad [4-\varepsilon, 4+\varepsilon] \subseteq [3,5].$$

Set

$$K = \sup_{(x,y)\in[-1,1]\times[3,5]} |g_y(x,y)|$$

and

$$M = \sup_{(x,y)\in[-1,1]\times[3,5]} |g(x,y)|.$$

Since g and g_y are continuous, these suprema are finite.

We now choose δ so that T maps Y into itself and is a contraction.

2021 Q4

For $y \in Y$,

$$\begin{aligned} \|T(y) - 4\|_\infty &= \sup_{|x| \le \delta} \left| \int_0^x g(t, y(t))\, dt \right| \\ &\le \sup_{|x| \le \delta} \int_0^{|x|} M\, dt \\ &\le \delta M. \end{aligned}$$

Therefore, it is enough to choose δ such that

$$\delta M < \varepsilon.$$

Then

$$|T(y)(x) - 4| < \varepsilon$$

for all $x \in [-\delta, \delta]$, so $T(y) \in Y$.

2021 Q4

For $y_1, y_2 \in Y$, by the Mean Value Theorem,

$$\begin{aligned}\|T(y_1) - T(y_2)\|_\infty &\leq \sup_{|x|\leq\delta} \int_0^{|x|} K|y_1(t) - y_2(t)|\,dt \\ &\leq K\delta\|y_1 - y_2\|_\infty.\end{aligned}$$

Thus T is a contraction provided

$$K\delta < 1.$$

So, given ε, choose δ such that

$$\delta < \frac{1}{K}, \qquad \delta < \frac{\varepsilon}{M}.$$

The Fixed Point Theorem then gives a fixed point $y \in Y$, which is the desired solution.

3. Sequences and Series of Functions

Pointwise Convergence

Definition

Let X be a set. A sequence of functions $f_n : X \to \mathbb{R}$ converges pointwise to a function $f : X \to \mathbb{R}$ if, for every $x \in X$ and every $\varepsilon > 0$, there exists $K(x, \varepsilon) \in \mathbb{N}$ such that

$$|f_n(x) - f(x)| < \varepsilon$$

whenever $n \geq K(x, \varepsilon)$.

Pointwise Convergence

Definition

Let X be a set. A sequence of functions $f_n : X \to \mathbb{R}$ converges pointwise to a function $f : X \to \mathbb{R}$ if, for every $x \in X$ and every $\varepsilon > 0$, there exists $K(x, \varepsilon) \in \mathbb{N}$ such that

$$|f_n(x) - f(x)| < \varepsilon$$

whenever $n \geq K(x, \varepsilon)$.

What this means is that the function values of f_n approach the function values of f as $n \to \infty$, that is $f_n(x) \to f(x)$, for all $x \in X$.

Uniform Convergence

Definition

Let X be a set. A sequence of functions $f_n : X \to \mathbb{R}$ converges uniformly to a function $f : X \to \mathbb{R}$ if, for every $\varepsilon > 0$, there exists $K(\varepsilon) \in \mathbb{N}$ such that

$$|f_n(x) - f(x)| < \varepsilon$$

for every $x \in X$, whenever $n \geq K(\varepsilon)$.

Uniform Convergence

We have another equivalent definition for uniform convergence.

Definition

We say f_n converges uniformly to f if for all $\varepsilon > 0$, there exists $K(\varepsilon) \in \mathbb{N}$ such that

$$\sup_{x \in X} |f_n(x) - f(x)| = d_\infty(f_n, f) < \varepsilon$$

whenever $n \geq K(\varepsilon)$.

Pointwise vs Uniform Convergence

Fact

Uniform convergence implies pointwise convergence, but pointwise convergence does not necessarily imply uniform convergence.

Pointwise vs Uniform Convergence

Fact

Uniform convergence implies pointwise convergence, but pointwise convergence does not necessarily imply uniform convergence.

Example

Consider the function $f_n : \mathbb{R} \to \mathbb{R}$ given by

$$f_n(x) = \begin{cases} 67 & x = n \\ 0 & \text{otherwise} \end{cases}.$$

Then clearly $\lim_{n\to\infty} f_n(x) = 0$ for all $x \in \mathbb{R}$, so f_n converges pointwise to 0, but $\sup_{x\in\mathbb{R}} |f_n(x) - 0| = 67$ for all $n \in \mathbb{N}$, so f_n does not converge uniformly to 0.

Useful Identities

Lemma

If $\lim_{m \to \infty} f_m(x) = f(x)$ and g is a continuous function, then

$$\lim_{m \to \infty} g(f_m(x)) = g(f(x))$$

for all x.

Useful Identities

Lemma

If $\lim_{m\to\infty} f_m(x) = f(x)$ and g is a continuous function, then

$$\lim_{m\to\infty} g(f_m(x)) = g(f(x))$$

for all x.

Proof. Fix x and let δ be given by the continuity of g. Since $\lim_{m\to\infty} f_m(x) = f(x)$, there exists $N(\delta) \in \mathbb{N}$ such that

$$|f_k(x) - f(x)| < \delta$$

for all $k \geq N$. By continuity of g, this implies

$$|g(f_k(x)) - g(f(x))| < \varepsilon.$$

Thus, we obtain the desired result □

Useful Identities

Lemma

If (f_n) is a Cauchy sequence in $C[a, b]$ under d_∞, then $(f_n(x))$ is a Cauchy sequence in $\mathbb{R}$ under $|\cdot|$.

Useful Identities

Lemma

If (f_n) is a Cauchy sequence in $C[a,b]$ under d_∞, then $(f_n(x))$ is a Cauchy sequence in $\mathbb{R}$ under $|\cdot|$.

Proof. Since (f_n) is Cauchy under d_∞, we have that

$$\sup_{x\in[a,b]} |f_n(x) - f_m(x)| < \varepsilon$$

whenever $n, m \geq K(\varepsilon) \in \mathbb{N}$. So, for all $x \in [a,b]$, we have that

$$|f_n(x) - f_m(x)| \leq \sup_{x\in[a,b]} |f_n(x) - f_m(x)| < \varepsilon$$

hence $(f_n(x))$ is Cauchy in $\mathbb{R}$. □

Useful Identities

Uniform Convergence Theorem

Suppose that (f_n) converges uniformly to f in $C[a,b]$. Then we have that

$$\lim_{n\to\infty}\int_a^b f_n(x)dx = \int_a^b f(x)dx.$$

Useful Identities

Uniform Convergence Theorem

Suppose that (f_n) converges uniformly to f in $C[a,b]$. Then we have that

$$\lim_{n\to\infty}\int_a^b f_n(x)dx = \int_a^b f(x)dx.$$

Proof. Since (f_n) converges uniformly to f, we know that for all $\varepsilon > 0$, there exists $N(\frac{\varepsilon}{b-a}) \in \mathbb{N}$ such that $\sup_{x\in[a,b]} |f_n(x) - f(x)| < \frac{\varepsilon}{b-a}$ whenever $n \geq N$. □

Useful Identities

So, whenever $n \geq N$, we have that

$$\begin{aligned}
\left|\int_a^b f_n(x)dx - \int_a^b f(x)dx\right| &= \left|\int_a^b f_n(x) - f(x)dx\right| \\
&\leq \int_a^b |f_n(x) - f(x)|dx \quad \leq \int_a^b \sup_{x\in[a,b]} |f_n(x) - f(\\
&= (b-a)\|f_n - f\|_\infty \\
&< \varepsilon
\end{aligned}$$

and so the limit is true.

L^p Spaces

Definition

Let $f : [a, b] \to \mathbb{R}$ be a Riemann-integrable function. Then, we define the L^p norm to be

$$\|f\|_p = \left(\int_a^b |f(x)|^p dx \right)^{\frac{1}{p}}.$$

L^p Spaces

Definition

Let $f : [a, b] \to \mathbb{R}$ be a Riemann-integrable function. Then, we define the L^p norm to be

$$\|f\|_p = \left(\int_a^b |f(x)|^p dx \right)^{\frac{1}{p}}.$$

Definition

We define L^p as the following:

$$L^p = \{f \in C[a, b] : \|f\|_p < \infty\}$$

L^p Spaces

Definition

We say that (f_n) converges to f in L^p if

$$\lim_{n\to\infty} \int_a^b |f_n(x) - f(x)|^p dx = 0.$$

Useful Identities: L^p Edition

Lemma

Let f be integrable over $[a, b]$, then for all p we have that

$$\|f\|_p \leq (b-a)^{\frac{1}{p}} \|f\|_\infty.$$

Hölder's Inequality

Hölder's Inequality for Integrals

Suppose f, g are integrable over an interval I and $p, q \in [1, \infty)$ satisfy $\frac{1}{p} + \frac{1}{q} = 1$. Then

$$\int_I |f(x)g(x)| dx \leq \left(\int_I |f(x)|^p \right)^{\frac{1}{p}} \left(\int_I |g(x)|^q \right)^{\frac{1}{q}}.$$

Relationships between L^p Spaces

Theorem (Week 5 Shortlist Q2)

Let (f_n) be a sequence in $C[a, b]$, and suppose the sequence converges to f in L^p. Then (f_n) converges to f in L^q, for all $1 \leq q \leq p$.

Relationships between L^p Spaces

Theorem (Week 5 Shortlist Q2)

Let (f_n) be a sequence in $C[a,b]$, and suppose the sequence converges to f in L^p. Then (f_n) converges to f in L^q, for all $1 \leq q \leq p$.

Proof. Since $1 \leq q \leq p$, we can use Hölder's Inequality with conjugates $\frac{p}{q}, \frac{p}{p-q}$

$$\int_a^b |f_n(x) - f(x)|^q dx \leq \left(\int_a^b (|f_n(x) - f(x)|^q)^{\frac{p}{q}} dx\right)^{\frac{q}{p}} \cdot \left(\int_a^b |1|^{\frac{p}{p-q}} dx\right)^{\frac{p-q}{p}}$$

$$= (b-a)^{\frac{p-q}{p}} \left(\int_a^b |f_n(x) - f(x)|^p dx\right)^{\frac{q}{p}}$$

and so

$$0 \leq \|f_n - f\|_q \leq (b-a)^{\frac{p-q}{pq}} \|f_n - f\|_p.$$

□

Series of Functions

Theorem

If a series $\sum_{n=0}^{\infty} x_n$ converges absolutely, that is $\sum_{n=0}^{\infty} |x_n|$ converges, then the series converges.

Series of Functions

Theorem

Let E be a Banach space with norm $\|\cdot\|$, and let (x_n) be a sequence in E. Then the series $\sum_{n=0}^{\infty} x_n$ converges in E if the series $\sum_{n=0}^{\infty} \|x_n\|$ converges in $\mathbb{R}$.

Weierstrass M-Test

Corollary: Weierstrass M-Test

Let (f_n) be a sequence of real-valued functions on a set X and suppose that there exists a real-valued sequence (M_n) such that

$$|f_n(x)| \leq M_n$$

for all $x \in X$ and $n \in \mathbb{N}$. If the sequence $\sum_{n=0}^{\infty} M_n$ converges, then the series $\sum_{n=0}^{\infty} f_n$ converges uniformly in X.

Convergence of Derivatives

Theorem (Week 5 Shortlist Q6)

Suppose $(f_n) \subseteq C^1[a,b]$ converge uniformly on $[a,b]$ to f and that the sequence (f_n') converges uniformly to g on $[a,b]$. Then $f \in C^1[a,b]$ and $f_n' \to f'$ uniformly on $[a,b]$.

Convergence of Derivatives

Proof. Since $f_n \in C^1$, by the Fundamental Theorem of Calculus, we have that

$$f_n(x) - f_n(y) = \int_y^x f_n'(t)dt.$$

Applying the limit as $n \to \infty$ to both sides, by Uniform Convergence Theorem, we get that

$$f(x) - f(y) = \int_y^x g(t)dt.$$

Applying the Fundamental Theorem of Calculus again, we have that

$$f'(x) = g(x)$$

for all $x \in [a, b]$, and so we proved it. Umazing! $\square$

Cauchy-Hadamard Theorem

Definition

Let $\sum_{n=0}^{\infty} a_n x^n$ be a power series. Then we call $b = \limsup_{n\to\infty} |a_n|^{\frac{1}{n}}$ the Cauchy-Hadamard parameter.

Cauchy-Hadamard Theorem

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Here, $\limsup_{n\to\infty}$ denotes $\lim_{n\to\infty} \sup_{k\geq n}$, that is, the supremum of the tail elements.

Cauchy-Hadamard Theorem

Theorem (Week 5 Shortlist Q4)

Suppose $\sum_{n=0}^{\infty} a_n x^n$ is a power series and $b = \limsup_{n\to\infty} |a_n|^{\frac{1}{n}}$. Then the power series converges if $b|x| < 1$ and diverges if $b|x| > 1$.

Cauchy-Hadamard Theorem

Theorem (Week 5 Shortlist Q4)

Suppose $\sum_{n=0}^{\infty} a_n x^n$ is a power series and $b = \limsup_{n\to\infty} |a_n|^{\frac{1}{n}}$. Then the power series converges if $b|x| < 1$ and diverges if $b|x| > 1$.

Equivalently, if we set $R = 1/b$, then the series converges whenever $|x| < R$. We call R the radius of convergence.

Cauchy-Hadamard Theorem

Proof. It suffices to check that the series converges absolutely. Let $b = \limsup_{n\to\infty} |a_n|^{\frac{1}{n}}$, then by definition, for all $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that $|a_n|^{\frac{1}{n}} < b + \varepsilon$ for all $n \geq N$. Now, we choose q such that $|x|b < q < 1$ and $\varepsilon = \frac{q}{|x|} - b > 0$, so now for all $n \geq N$ we have that $|a_n|^{\frac{1}{n}} < \frac{q}{|x|}$, which is equivalent to $|a_n|^{\frac{1}{n}}|x| < q < 1$. $\square$

Cauchy-Hadamard Theorem

As a result, we have that

$$\begin{aligned}
\sum_{n=0}^{\infty} |a_n x^n| &= \sum_{n=0}^{\infty} (|a_n|^{\frac{1}{n}} |x|)^n \\
&= \sum_{n=0}^{N-1} (|a_n|^{\frac{1}{n}} |x|)^n + \sum_{n=N}^{\infty} (|a_n|^{\frac{1}{n}} |x|)^n \\
&< \sum_{n=0}^{N-1} (|a_n|^{\frac{1}{n}} |x|)^n + \sum_{n=N}^{\infty} q^n \\
&= \sum_{n=0}^{N-1} (|a_n|^{\frac{1}{n}} |x|)^n + \frac{q^N}{1-q} \\
&< +\infty.
\end{aligned}$$

Differentiation of Power Series

Theorem

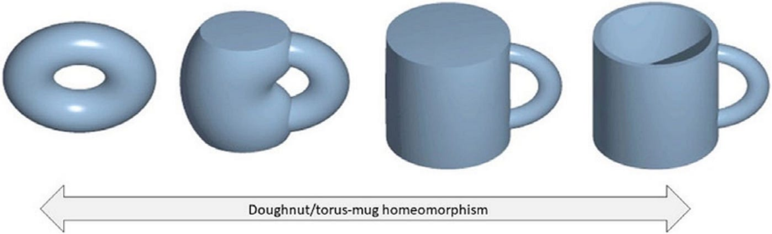
Suppose $\sum_{n=0}^{\infty} a_n x^n$ with radius of convergence $R > 0$. Then the series is differentiable on $(-R, R)$ and

$$\frac{d}{dx} \sum_{n=0}^{\infty} a_n x^n = \sum_{n=1}^{\infty} n a_n x^{n-1}.$$

4. Topological Spaces

What is a topology?

What is a topology?



Topological Spaces

Definition

Let X be a non-empty set. A family of subsets of X $\tau \subseteq \mathcal{P}(X)$ is called a *topology* if

1. $\emptyset, X \in \tau$;
2. If $\{V_i\}_{i \in I} \subseteq \tau$, then $\bigcup_{i \in I} V_i \in \tau$;
3. If $V_1, V_2 \in \tau$, then $V_1 \cap V_2 \in \tau$

We call the pair (X, τ) a topological space, and the sets in τ open sets.

Topological Spaces – Important Topologies

Examples of topologies

- *Discrete topology*: $\tau = \mathcal{P}(X)$.
- *Coarse topology*: $\tau = \{\emptyset, X\}$. Also known as the *indiscrete topology*.
- *Metric topology*: Let (X, d) be a metric space. Then (X, d) induces a topology $\tau_d = \{V \subseteq X : V = \text{int}(V)\}$.
- *Cofinite topology*: $\tau = \{\emptyset\} \cup \{U \subseteq X : X \setminus U \text{ is finite}\}$.
- *Cocountable topology*: $\{\emptyset\} \cup \{U \subseteq X : X \setminus U \text{ is countable}\}$.
- *Subspace topology*: Let (X, τ) be a topological space and consider a subset $S \subseteq X$. Then

$$\tau \,|_S = \{S \cap U : U \in \tau\}$$

forms a topology on S.

Closed Sets

Just like before in metric spaces, we have a notion of closed sets in topological spaces.

Definition

Let (X, τ) be a topological space, A subset $Y \subseteq X$ is called closed if Y^c is open.

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Definition / Theorem

Let $\mathcal{C}(X)$ denote the collection of closed subsets of X. Then the elements of $\mathcal{C}(X)$ satisfy:

- $\emptyset, X \in \mathcal{C}(X)$
- If $(V_i)_{i \in I} \subseteq \mathcal{C}(X)$, then $\cap_{i \in I} V_i \in \mathcal{C}(X)$
- If $V_1, V_2 \in \mathcal{C}(X)$, then $V_1 \cup V_2 \in \mathcal{C}(X)$

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The above follows from De Morgan's Laws.

Neighbourhoods

Definition

Let (X, τ) be a topological space. If $x \in X$ then we say a set $V \in \tau$ is an open neighbourhood of x if $x \in V$. A neighbourhood of x is any subset of X containing an open neighbourhood of x. For the collection of neighbourhoods of x, we write $\mathrm{Nbhd}(x)$.

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Let $Y \subseteq X$. We define the interior of Y to be

$$\mathrm{Int}(Y) = \{y \in Y : \exists V \in \mathrm{Nbhd}(y), V \subseteq Y\}.$$

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Definition

Let $Y \subseteq X$. We define the interior of Y to be

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We are essentially replacing open balls with open neighbourhoods.

Boundary and Closure

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If $Y \subseteq X$ then the boundary of Y is

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Definition

We define the closure of $Y \subseteq X$ to be

$$\mathrm{cl}(Y) = \mathrm{Int}(Y) \cup \mathrm{Bd}(Y).$$

Convergence in Topological Spaces

Definition

Let (X, τ) be a topological space. A sequence $(x_n) \subseteq X$ converges to $x \in X$ if for every $V \in \text{Nbhd}(x)$ there exists $K(V) \in \mathbb{N}$ such that

$$x_n \in V, \quad \forall n \geq K(V).$$

Again, we are generalising the notion of convergence by replacing open balls with neighbourshoods.

Example on Convergence

Week 7 Shortlist Q1

Let X be a set. Recall that the cofinite topology on X is defined by

$$\tau = \{\emptyset\} \cup \{U \subseteq X : X \backslash U \text{ is finite}\}.$$

Now, let $X = \mathbb{N}$ be equipped with the cofinite topology.

i) Find all limits of the sequence (n) or prove that no limits exist.
ii) Find all limits of the sequence $(1 + (-1)^n)$ or prove that no limits exist.
iii) Characterise all convergent sequences.

Example on Convergence

Before we think about convergence, we should think about what the neighbourhoods of $x \in X$ look like. We can see that τ can be equivalently defined as

$$\tau = \{\emptyset\} \cup \{X \backslash V : V \subseteq X \text{ is finite}\},$$

so in the case where $X = \mathbb{N}$, the neighbourhoods of x are the sets which can be written as the difference between X and a finite subset of X that still contain x, that is,

$$\mathrm{Nbhd}(x) = \{X \backslash V : V \subseteq X \text{ is finite, } x \in X \backslash V\}.$$

Example on Convergence

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Proof. Let $x \in \mathbb{N}$ and recall that

$$\text{Nbhd}(x) = \{X \backslash V : V \subseteq X \text{ is finite, } x \in X \backslash V\}.$$

Suppose $V \in \text{Nbhd}(x)$, then $X \backslash V$ is a finite set, so write $\{a_1, ..., a_k\}$ for $X \backslash V$ in ascending order. Then $V = X \backslash \{a_1, ..., a_k\}$, so if $K(V) = a_k + 1$ then clearly we have that

$$\{a_k + 1, a_k + 2, ...\} \subseteq V$$

which implies that (n) converges to everything in $\mathbb{N}$. □

Example on Convergence

Part ii)

Find all the limits of the sequence $(1 + (-1)^n)$ or prove that no limits exist.

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Example on Convergence

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Claim: No limits exist.

Proof. Suppose the sequence converges to 0. Then we can pick $V = X \setminus \{2\} \in \text{Nbhd}(0)$ such that $1 + (-1)^n = 0 \in V$, for all $n \geq K(V)$. But $1 + (-1)^{n+1} = 2 \notin V$ so the sequence cannot converge to 0. In a similar way, we can show that the sequence cannot converge to 2 or any other element in $\mathbb{N}$, so no limit exists. $\square$

Example on Convergence

Part iii)

Characterise all convergent sequences (in (X, τ)).

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Claim: Convergent sequences are either eventually constant, which the limit is the constant value, or repeats values a finite number of times, in which case the limit is any number in $\mathbb{N}$.

Proof. If a sequence (a_n) is eventually constant at value x, then for any $V \in \mathrm{Nbhd}(x)$ take $K(V)$ to be the index such that $a_n = x$ for all $n \geq K(V)$ (where the sequence becomes constant). Then clearly $a_n = x \in V$ for all $n \geq K(V)$ and $V \in \mathrm{Nbhd}(x)$.

Example on Convergence

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Proof. If a sequence (a_n) is eventually constant at value x, then for any $V \in \text{Nbhd}(x)$ take $K(V)$ to be the index such that $a_n = x$ for all $n \geq K(V)$ (where the sequence becomes constant). Then clearly $a_n = x \in V$ for all $n \geq K(V)$ and $V \in \text{Nbhd}(x)$. If (a_n) repeats values a finite number of times, then let $x \in \mathbb{N}$ and consider $V \in \text{Nbhd}(x)$. Then we can write $V = X \backslash \{b_1, ..., b_k\}$, as $V \in \tau$. Since (a_n) repeats numbers only a finite number of times, there exists $K(V)$ such that $a_n \notin \{b_1, ..., b_k\}$ for all $n \geq K(V)$, so $a_n \in V$. Since V and x are arbitrary, the limit is any number in $\mathbb{N}$. □

Example on Convergence

Now suppose the sequence repeats a value infinitely many times, then the limit does not exist as we can take a neighbourhood that includes any candidate for a limit but excludes the repeated value, so there cannot be an index such that $a_n \in V$ always for sufficiently large n.

Continuity on Topological Spaces

Our next piece of machinery is to make sense of what continuity of functions look like on topological spaces.

On metric spaces (X, d_X) and (Y, d_Y), we said that a function $f : X \to Y$ was continuous if the preimage of each open set $U \in (Y, d_Y)$ is also an open set; that is, for each open set $U \in (Y, d_Y)$, $f^{-1}(U)$ is also an open set in (X, d_X).

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We can generalise this by observing that each open set of X, where (X, τ) is now a topological space, belongs to τ. Therefore, the natural generalisation goes as follows:

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We can generalise this by observing that each open set of X, where (X, τ) is now a topological space, belongs to τ. Therefore, the natural generalisation goes as follows:

Definition

Let (X, τ_X) and (Y, τ_Y) be topological spaces. A function $f : X \to Y$ is continuous if, for every $U \in \tau_Y$, we have that $f^{-1}(U) \in \tau_X$.

2022, Q8

(2022, Problem 8)

Suppose that (X_1, τ_1) and (X_2, τ_2) are topological spaces and that $f : X_1 \to X_2$ is continuous. For each of the following statements, say whether it must be true. If that is not the case, give a counterexample.

i) $f(U) \in \tau_2$ for all $U \in \tau_1$.

ii) $f^{-1}(V) \in \tau_1$ for all $V \in \tau_2$.

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Remark. Continuity does not tell you information about the *image* of an open set unless the subset is compact.

Hausdorff Topological Spaces

Let (X, τ) be a topological space. In the previous slides, neighbourhoods captured the concept of "closeness" for two points in the space X under the topology τ.

- We still need a notion of "farness".
- What sort of definition would allow us to satisfactorily say that two points are "far away"?

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- We still need a notion of "farness".
- What sort of definition would allow us to satisfactorily say that two points are "far away"?

Definition

- On a metric space, if x and y are not the same point, then a metric should ensure that $d(x, y) > 0$.
- How do we define this on a topology?

If x, y are not the same point, then we can always find a pair of open sets in τ that do not intersect!

Hausdorff Topological Spaces

Definition

Let (X, τ) be a topological space. We say that (X, τ) is *Hausdorff* if, for all $x, y \in X$ such that $x \neq y$, there exist $U, V \in \tau$ such that $x \in U$, $y \in V$, and $U \cap V = \emptyset$.

Limits in Hausdorff Spaces

If a topological space is Hausdorff, limits of sequences on the space behave nicely! We obtain the uniqueness of limits property that we come to expect.

(2018, Problem 8)

Let (X, τ) be a topological space. Let (X', τ') be a Hausdorff topological space, and suppose that

$$f : X \to X'$$

is a continuous, injective function. Prove that X is a Hausdorff space as well.

2018, Q8

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is a continuous, injective function. Prove that X is a Hausdorff space as well.

Proof. Consider $x, y \in X, x \neq y$. Clearly $f(x) \neq f(y)$ as f injective; there exists open $U \ni f(x), V \ni f(y)$ such that $U \cap V = \emptyset$, clearly $f^{-1}(U), f^{-1}(V)$ work. $\square$

Bases for Topologies

There are many ways to define a topology on the same space.
Oftentimes, we want a simple description of the topology.

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For example, we note an open set of $\mathbb{R}$ with the usual topology can be annoying to describe, like so

$$(0,1) \cup \left(1, \frac{6}{8}\right) \cup (2067, 2420).$$

But one can note this isn't that much different from our usual open intervals. All we've done is "union-ed" a bunch.

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But one can note this isn't that much different from our usual open intervals. All we've done is "union-ed" a bunch.

Definition

Let (X, τ) be a topological space.
A *base* for τ is a subset $\mathcal{B} \subseteq \tau$ such that every open set $V \in \tau$ can be expressed as a union of elements in $\mathcal{B}$; that is, for every $V \in \tau$, one can express V as

$$V = \bigcup_{i \in I} V_i, \text{ where } V_i \in \mathcal{B} \ \ \forall i \in I.$$

Basis of $\mathbb{R}$

The open intervals form a basis of $\mathbb{R}$.

We also have a notion of a base at a point $x \in X$. One can think of this as a localisation of the notion of a base of a topology; instead of looking at the entire space, one can define a topology at a point.

Definition

Let (X, τ) be a topological space, and consider a point $x \in X$. A *local base* for τ at x is a collection $\mathcal{LB}_x \subseteq \tau$ of open neighbourhoods of x such that if U is any neighbourhood of x, then there exist some $V \in \mathcal{LB}_x$ such that $V \subseteq U$.

Bases for Topologies

As always, there are different ways to characterise a *base* $\mathcal{B}$. Recall that a base for a topology τ can be thought of as the *smallest* set of open sets such that the union encapsulates all of the open sets in X.
We must, therefore, require a few properties to hold:

- $\mathcal{B}$ must cover all of the elements in X.
  - This is because X is an open set in X, so we must be able to cover X.
- $\mathcal{B}$ should be as "small" as possible, in the sense that, if $x \in V_1 \cap V_2$ for $V_1, V_2 \in \mathcal{B}$, then we can always find some other open set $V \in \mathcal{B}$ such that $x \in V$.

It turns out that these are enough to define a base of a topology! We state this in more formal language.

Bases for Topologies

Theorem

Let X be a set, and let $\mathcal{B} \subseteq \mathcal{P}(X)$ be a collection of subsets. Then

$$\tau = \{V \subseteq X : V \text{ is a union of sets in } \mathcal{B}\}$$

is a topology if and only if the following properties are satisfied:

- $\bigcup_{V \in \mathcal{B}} V = X$; in other words, $\mathcal{B}$ *covers* X.
- For every V_1 and V_2 in $\mathcal{B}$ and every $x \in V_1 \cap V_2$, there exist a $V \in \mathcal{B}$ such that
$$x \in V \subseteq V_1 \cap V_2.$$

- The collection union of sets in $\mathcal{B}$ forms a topology with base $\mathcal{B}$.

2022, Q6

(2022, Problem 6)

- Give the definition of a base for a topology τ on a set X.
- Let $\mathcal{G}$ consist of the empty set and the collection of all open rectangles in the plane. That is,

$$\mathcal{G} = \{\emptyset\} \cup \{(a, b) \times (c, d) \subseteq \mathbb{R}^2 : a < b,\, c < d\}.$$

Prove that $\mathcal{G}$ is a base for a topology on $\mathbb{R}^2$.

2022, Q6

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Prove that $\mathcal{G}$ is a base for a topology on $\mathbb{R}^2$.

Proof. We use the above criterion; first condition is true as for every (x, y), we can take $(x-1, x+1) \times (y-1, y+1)$. For second condition, if we have $p \in ((a,b) \times (c,d)) \cap ((a', b') \times (c', d'))$, take $(\max(a, a'), \min(b, b')) \times ((\max(c, c') \times \min(d, d'))$. We also note one could have directly proved it is a basis in the standard topology. $\square$

Bases for Topologies

It turns out that the collection of open sets generated by taking all finite intersections is also a topology! Specifically, if $S \subseteq \mathcal{P}(X)$ is *any* collection of subsets of X, then the collection

$$\mathcal{B} = \{V_1 \cap \cdots \cap V_n : V_k \in S, \quad k = 1, \ldots, n\}$$

forms a base of the same topology

$$\tau = \{V \subseteq X : V \text{ is a union of sets in } \mathcal{B}\}.$$

In particular, S *generates* τ. We say that S is a *subbase* and we often write τ as $\tau(S)$ to mean that S generates the topology τ; that is, we also have that

$$\tau(S) = \{V \subseteq X : V \text{ is a union of sets in } \mathcal{B}\}.$$

We make this definition formal.

Bases for Topologies

Definition

Let X be a set, and let $S \subseteq \mathcal{P}(X)$ be *any* collection of subsets. Define $\mathcal{B}$ to be the set of all **finite** intersections of sets in S; that is,

$$\mathcal{B} = \{V_1 \cap \cdots \cap V_n : V_k \in S, \quad k = 1, \ldots, n\}.$$

Then $\mathcal{B}$ forms a base of the topology

$$\tau(S) = \{V \subseteq X : V \text{ is a union of sets in } \mathcal{B}\}.$$

We call S a *subbase* for $\tau(S)$ and say that τ is generated by S.

- We remark that we also allow the empty intersection; by convention, this gives the entire space X.

Convergence in topological spaces

- We talked about convergence on a topological space previously with a sequence of points. What about a sequence of functions?
- We can mimic the same ideas.

Remember that a sequence of points $\{x_n\}_{n=1}^{\infty}$ converge to a point $x \in X$ if, for each open neighbourhood of x, all *sufficiently large* points in the sequence belong to the neighbourhood.

- We can mimic this idea but we need to be a little careful with what we mean by "*sufficiently large points*" because functions are not points.

Pointwise convergence

Definition

Let X be a set and $Y = F(X, \mathbb{R})$ be the set of real-valued functions on X. For each $x \in X, y \in \mathbb{R}$, and $\varepsilon > 0$, define

$$V_{x,y,\varepsilon} = \{g \in Y : |g(x) - y| < \varepsilon\}.$$

Then let

$$S = \{V_{x,y,\varepsilon}\}_{x \in X, y \in \mathbb{R}, \varepsilon > 0}$$

forms a topology $\tau_{pt} = \tau(S)$ generated by S as a subbase; we call this the *topology of pointwise convergence*.

- The topology of pointwise convergence on a set X forms a Hausdorff topological space.
- A sequence $\{f_n\}_{n=1}^{\infty}$, where $f_n : X \to \mathbb{R}$ converges *pointwise* to f if and only if $f_n \to f$ in the topology τ_{pt}.

Weak convergence

Definition

Let H be a Hilbert space. We say that a sequence of vectors $\{\mathbf{x}_n\}_{n=1}^{\infty}$ converges *weakly* to a vector $\mathbf{x} \in \mathsf{H}$ if, for each vector $y \in \mathsf{H}$, we have that

$$\langle \mathbf{x}_n, \mathbf{y} \rangle \to \langle \mathbf{x}, \mathbf{y} \rangle.$$

This allows us to define a notion of a *weak topology* in a similar light. Notice that this requirement is equivalent to ensuring that

$$\langle \mathbf{x}_n - \mathbf{x}, y \rangle \to 0.$$

To make sense of this on a topological space, we construct the topology

$$V_{\mathbf{x},\mathbf{y},\varepsilon} = \{\mathbf{z} : |\langle \mathbf{z} - \mathbf{x}, \mathbf{y} \rangle| < \varepsilon\}.$$

This gives the *weak topology* $\tau_{weak} = \tau(S)$ generated by the subbase

$$S = \{V_{\mathbf{x},\mathbf{y},\varepsilon}\}_{\mathbf{x} \in \mathsf{H}, \mathbf{y} \in \mathbb{R}, \varepsilon > 0}.$$

Week 8, Q1

Week 8, Problem 1

Prove that a sequence $(x_n) \subseteq H$ converges weakly to $x \in H$ if and only if it converges to x in the topology $\tau(S)$.

Week 8, Q1

Week 8, Problem 1

Prove that a sequence $(x_n) \subseteq H$ converges weakly to $x \in H$ if and only if it converges to x in the topology $\tau(S)$.

Proof. For reverse direction, we have by definition it follows. For forward direction, consider $x \in \bigcap_{i=1}^{k} V_{x_i,y_i,\varepsilon_i} \subseteq U$. As $x \in V_{x_i,y_i,\varepsilon_i}$, consider $\gamma_i = \varepsilon_i - |\langle x - x_i, y_i\rangle| > 0$, we have using weak convergence, there is some N_i s.t $|\langle x_n - x, y_i\rangle| < \gamma_i$, i.e., $x_n \in V_{x,y_i,\gamma_i} \subseteq V_{x_i,y_i,\varepsilon_i}$ for all $n \geq N_i$. □

Nets

Definition

A *directed set* is a set Λ, together with a binary relation $\leq$ satisfying the following properties, for all $i, j, k \in \Lambda$.

- $i \leq i$ (i.e. $(\Lambda, \leq)$ is reflexive).
- $i \leq j$, $j \leq k$ implies that $i \leq k$ (i.e. $(\Lambda, \leq)$ is transitive).
- There exists some $m \in \Lambda$ such that, for all pairs $i, j \in \Lambda$, we have that $i, j \leq m$.
  - This is the *upper bound property* for pairs of elements in Λ.

- Like how a sequence $\{x_k\}_{k \in \mathbb{N}}$ can be viewed as a function from $\mathbb{N}$ to some set X, we can view sequences $\{x_\lambda\}_{\lambda \in \Lambda}$ as a function from the directed set Λ to some set X.
- The way we map an element $\lambda \in \Lambda$ from Λ to the set X is called a *net*.

Nets

Definition

A *net* is a function from a directed set $(\Lambda, \leq)$ to a set X.

With this in mind, how can we meaningfully talk about convergence of a net?

- Remember that a net comes from a *directed* set, so the same idea to how we normally think of convergence applies.

Definition

Let (X, τ) be a topological space. A net $\{x_\lambda\}_{\lambda \in \Lambda}$ converges to a point $x \in X$ if, for every neighbourhood V of x, there exist an $\alpha(V) \in \Lambda$ such that, for all $\lambda \geq \alpha$, we have that $x_\lambda \in V$.

Homeomorphisms

One big question of interest is when two spaces are more-or-less *the same*.

- The classic example is that we can always deform a coffee mug into a donut and vice versa without tearing or cutting the surface.

That is, given topological spaces (X, τ_X) and (Y, τ_Y), when are these *the same*? We note that open sets essentially retain all the information about topological spaces. So really what we want is a bijection between two spaces which preserves open sets in both directions.

Definition

Let (X, τ_X) and (Y, τ_Y) be topological spaces. A bijection $f : X \to Y$ is called a *homeomorphism* if f and f^{-1} are continuous. We say that X and Y are *homeomorphic* if there exists a homeomorphism $f : X \to Y$.

Homeomorphic invariants

Definition

A topological space (X, τ) is *connected* if it is not the union of two disjoint nonempty open subsets.

Motivating problem

Let X be a topological space. Show that there exists a nonconstant continuous function $X \to \{0, 1\}$ if and only if X is disconnected

The nice thing about connected spaces is that it is invariant under continuous functions; that is, if (X, τ_X) and (Y, τ_Y) are topological spaces where (X, τ_X) is connected, then $f(X) \subseteq Y$ is also connected.

Theorem

Let (X, τ_X) and (Y, τ_Y) be topological spaces, and suppose that $f : X \to Y$ is a continuous function. If X is connected, then so if $f(X)$.

Homeomorphic invariants

Definition

A topological space (X, τ) is *path-connected* if, for every pair of points $x, y \in X$, there is a continuous function $f : [0, 1] \to X$ such that $f(0) = x$ and $f(1) = y$.

- Path-connectedness implies connectedness.

(2018, Problem 7)

(a) Prove that $X = \mathbb{R} \setminus \{0\}$ and $Y = \mathbb{R}^2 \setminus \{(0,0)\}$ are not homeomorphic.

(b) Prove that $\mathbb{R}$ and $\mathbb{R}^2$ are not homeomorphic.

(2018, Problem 7)

(a) Prove that $X = \mathbb{R} \setminus \{0\}$ and $Y = \mathbb{R}^2 \setminus \{(0,0)\}$ are not homeomorphic.

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Proof. Y is connected (because it is path-connected), while X is not. Let $f : \mathbb{R} \to \mathbb{R}^2$ be a homeomorphism and consider the restriction $f : \mathbb{R} \setminus \{f^{-1}(0,0)\} \to \mathbb{R}^2 \setminus \{(0,0)\}$, where the former is clearly homeomorphic to $\mathbb{R} \setminus \{0\}$. □

5. Compactness

Compactness

- In some sense, compactness is a way to deal with the notion of *finiteness*.
- On a topological space (X, τ), we can look at open sets that cover the entire space X. Ordinarily, we do not concern ourselves with the number of open sets required to cover X, just that it covers X.
  - But if we want to deal with *finiteness*, then one satisfying solution is that anytime we cover the entire space, we only want to do so with a finite number of open sets.

This gives us a concrete way to define *finiteness* on a topological space.

Compactness

Definition

A topological space (X, τ) is *compact* if, for every $\{V_i\}_{i \in I} \subseteq \tau$ such that

$$\bigcup_{i \in I} V_i = X,$$

there is a *finite* subset $\{i_1, \ldots, i_n\} \subseteq I$ such that

$$\bigcup_{k=1}^{n} V_{i_k} = X.$$

A subset $Y \subseteq X$ is compact iff it is compact in the subspace topology. So use $Y \subseteq \bigcup_{i \in I} V_i$ for definitions instead.

In other words, every cover has a *finite subcover*.

- For example, the interval $[0, 1]$ is compact.

Compactness

Compactness properties

1. Let (X, τ) be a compact topological space. If $Y \subseteq X$ is a closed subset of X, then Y is compact. (As Y^c can be added to an open cover for Y to make an open cover for X.)
2. Let (X, τ) be a Hausdorff topological space. If $Y \subseteq X$ is a compact subset of X, then Y is closed. (For each $y \in Y^c$, consider pairs of disjoint neighbourhoods of elements of Y and y.)

2018, Q3

(2018, Problem 3)

Let (X, τ) and (X', τ') be topological spaces.

(a) Prove that if $Y \subseteq X$ is compact and $f : X \to X'$ is continuous, then $f(Y)$ is a compact subset of X'.

(b) Suppose that Y' is a compact subset of X' and $f : X \to X'$ is continuous. Is $f^{-1}(Y')$ necessarily compact?

2018, Q3

(2018, Problem 3)

Let (X, τ) and (X', τ') be topological spaces.

(a) Prove that if $Y \subseteq X$ is compact and $f : X \to X'$ is continuous, then $f(Y)$ is a compact subset of X'.

(b) Suppose that Y' is a compact subset of X' and $f : X \to X'$ is continuous. Is $f^{-1}(Y')$ necessarily compact?

Proof. Consider any open cover $\{V_i\}_{i \in I}$ of $f(Y)$. For each $V_i \in \tau'$, let $U_i = f^{-1}(V_i) \in \tau$. U_i open as f is continuous, hence $\{U_i\}_{i \in I}$ form an open cover of Y. However, since Y is compact, each open cover of Y has a finite subcover $\{U_k\}_{k \in [1,n]}$. The images of the finite subcover $\{V_k\}_{k \in [1,n]}$ form a finite subcover of $f(Y)$; therefore, $f(Y)$ is also compact. False for part (b); consider $f : \mathbb{R} \to \mathbb{R} : x \mapsto 0$. Then $f^{-1}(\{0\}) = \mathbb{R}$ which is not bounded, hence not compact. $\square$

2020, Q7

(2020, Problem 7)

Let (X, τ_X) be a compact topological space and let (Y, τ_Y) be a Hausdorff topological space.

1. Show that if $f : X \to Y$ is a continuous map and $Z \subseteq X$ is closed, then $f(Z) \subseteq Y$ is closed.
2. Show that if $f : X \to Y$ is a continuous bijection, then it is a homeomorphism.

2020, Q7

(2020, Problem 7)

Let (X, τ_X) be a compact topological space and let (Y, τ_Y) be a Hausdorff topological space.

1. Show that if $f : X \to Y$ is a continuous map and $Z \subseteq X$ is closed, then $f(Z) \subseteq Y$ is closed.
2. Show that if $f : X \to Y$ is a continuous bijection, then it is a homeomorphism.

Proof. X is compact and $Z \subseteq X$ is closed; thus Z is compact. Since f is continuous, it follows that $f(Z) \subseteq Y$ is compact. But Y is Hausdorff; therefore, $f(Z)$ is closed.
Let $g = f^{-1}$. RTS g is continuous, i.e., if Z is closed in (X, τ_X), then $g^{-1}(Z)$ is closed in (Y, τ_Y). $f(Z)$ is closed in (Y, τ_Y) by (i), but $f(Z) = g^{-1}(Z)$; therefore, $g^{-1}(Z)$ is closed which implies that g is continuous. $\square$

Compactness

Heine–Borel Theorem

A subset $X \subseteq \mathbb{R}^n$ is compact if and only if X is closed and bounded.

Bolzano–Weierstrass Theorem

Every bounded sequence of real numbers has a convergent subsequence.

Bolzano–Weierstrass easily extends to show every bounded sequence in $\mathbb{R}^n$ has a convergent subsequence. (Apply to first coordinate, then again to second, and so on.)

2022, Q10

(2022, Problem 10)

Let $X_1 = \{z \in \mathbb{C} \mid |z| = 1\}$ and $X_2 = \{(x, y) \in \mathbb{R}^2 \mid x^2 - y^2 = 1\}$, both with the usual metric topology inherited from the plane. Prove X_1 is not homeomorphic to X_2.

2022, Q10

(2022, Problem 10)

Let $X_1 = \{z \in \mathbb{C} \mid |z| = 1\}$ and $X_2 = \{(x, y) \in \mathbb{R}^2 \mid x^2 - y^2 = 1\}$, both with the usual metric topology inherited from the plane. Prove X_1 is not homeomorphic to X_2.

Proof. We have X_1 is closed and bounded in the usual metric topology, and hence by Heine-Borel, it is compact. X_2 is unbounded and hence it is not compact. Since homeomorphisms preserve compactness, these spaces can not be homeomorphic. $\square$

Basic Compactness Results

Sequentially Compact

A topological space (X, τ) is *sequentially compact* if every sequence in X has a convergent subsequence.

Sequentially Compactness Characterization

Let $X \subseteq \mathbb{R}^n$. The following are equivalent.

- X is compact.
- X is sequentially compact.
- X is closed and bounded.

Continuity over compact spaces

Theorem

Let (X, τ_X) and (Y, τ_Y) be topological spaces. If $f : X \to Y$ is continuous and X is compact, then $f(X) \subseteq Y$ is compact.

We saw that $[0, 1]$ was compact.

- More generally, $[a, b] \subseteq \mathbb{R}$ is also compact.

Therefore, if we take a continuous function $f : X \to \mathbb{R}$ with $X = [a, b]$, then $f(X)$ is compact (and thus, closed and bounded on $\mathbb{R}$). This implies the max-min theorem.

Extreme Value (Max-Min) Theorem

Let $f : [a, b] \to \mathbb{R}$ be a continuous function. Then f attains maximum and minimum values.

Continuity over compact spaces

Recall that a function $f : X \to Y$ over metric spaces (X, d_X) and (Y, d_Y) is *uniformly continuous* if, for all $\varepsilon > 0$, there exists some $\delta(\varepsilon)$ such that

$$d_X(x, x') < \delta \implies d_Y(f(x'), f(x)) < \varepsilon.$$

Heine–Cantor Theorem

Let (X, d) be a compact metric space, and let $f : X \to \mathbb{R}$ be a continuous function. Then f is *uniformly continuous.* (Generalises exactly to any metric space.)

Compactness on metric spaces

We consider a metric space (X, d). We can talk about compactness on metric spaces by discussing concepts of boundedness.

Definition

A metric space (X, d) is said to be *totally bounded* if, for every $\varepsilon > 0$, there is a *finite* set $\{x_1, \ldots, x_n\} \subseteq X$ such that

$$X = \bigcup_{k=1}^{n} B(x_k, \varepsilon).$$

- If $Y \subseteq X$ is totally bounded, then *every* subset of Y is also totally bounded.
- Similarly, the closure of Y is also totally bounded.

Compactness on a metric space

Compact Metric Space Characterization

Let (X, d) be a metric space. The following are equivalent.

1. X is compact.
2. X is sequentially compact.
3. X is complete and totally bounded.

- In other words, every compact metric space is *complete*.
- However, not all complete metric spaces are *compact*. Euclidean space is complete, but neither compact nor bounded (and hence not totally bounded either).
- Complete metric spaces require total boundedness for it to be compact.

Equicontinuity

Definition

Let (X, d_X) and (Y, d_Y) be metric spaces. A subset S of $C(X, Y)$ is said to be

(1) Pointwise equicontinuous if, for every point $x \in X$ and $\varepsilon > 0$, there exists $\delta(x, \varepsilon)$ such that, for every $f \in S$, we have that

$$d_Y(f(x'), f(x)) < \varepsilon \quad \text{whenever } d_X(x', x) < \delta.$$

(2) Uniformly equicontinuous if, for every $\varepsilon > 0$, there exists $\delta(\varepsilon)$ such that, for every $f \in S$, we have that

$$d_Y(f(x'), f(x)) < \varepsilon \quad \text{whenever } d_X(x', x) < \delta.$$

Note that *pointwise* depends on the point and threshold ε, whereas *uniform* must hold for every point in X.

Arzela-Ascoli Theorem

Clearly, we see that uniform equicontinuity implies pointwise equicontinuity. When does pointwise equicontinuity imply uniform equicontinuity?

- When X is compact!

Theorem

Let (X, d_X) and (Y, d_Y) be metric spaces, with X compact. Then $S \subseteq C(X, Y)$ is pointwise equicontinuous if and only if S is uniform continuous.

Arzela-Ascoli Theorem

This previous theorem gives rise to an important theorem, the *Arzela-Ascoli Theorem*.

Arzela-Ascoli Theorem

A bounded subset of $(C[0,1], \|\cdot\|_\infty)$ is totally bounded if and only if it is equicontinuous.

- More generally, $[0,1]$ can be replaced with any compact metric space.
- The Arzela-Ascoli Theorem implies that any subset of $(C[0,1], \|\cdot\|_\infty)$ is compact if and only if it is closed, bounded, and equicontinuous.

The Weierstrass Approximation Theorem

We continue to look at the metric space $C[a,b]$ endowed with the $\|\cdot\|_\infty$ metric.

Weierstrass Approximation Theorem

Let f be a continuous function on a *closed* and *bounded* interval $[a,b]$. For any $\varepsilon > 0$, there exists a polynomial $p(x)$ such that

$$\|f - p\|_\infty < \varepsilon.$$

2022, Q9

(2022, Problem 9)

Let

$$V = \left\{ f \in C[0,1] : \int_0^1 f(t)\,dt = 0 \right\}$$

and let P denote the set of polynomials which lie in V.

(a) Give an example of a nonzero element of P.

(b) What is the closure of P in $(V, \|\cdot\|_\infty)$?

(c) Is $(V, \|\cdot\|_\infty)$ complete?

2022, Q9

(a) Let's just take $f(x) = x - \frac{1}{2}$, which will obviously integrate to 0 on this interval.

(b) It is sensible to suspect that the closure of P will be V, because of our approximation theorem. We note that $C[0,1]$ is the set of continuous functions on the closed and bounded interval $[0,1]$. Therefore, any function $f \in V$ has a polynomial $p(x)$ such that

$$\|f - p\|_\infty < \varepsilon/2.$$

Sadly, we don't get $p \in V$, but let's see how far it's integral is from 0.

2022, Q9

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Sadly, we don't get $p \in V$, but let's see how far it's integral is from 0.

$$\begin{aligned}\left|\int_0^1 p(x)\,dx\right| &= \left|\int_0^1 p(x) - f(x) + f(x)\,dx\right| \\ &\leq \left|\int_0^1 p(x) - f(x)\,dx\right| + \left|\int_0^1 f(x)\,dx\right| \\ &\leq \int_0^1 |p(x) - f(x)|\,dx \leq \|f - p\|_\infty < \frac{\varepsilon}{2}.\end{aligned}$$

2022, Q9

Let $C = \int_0^1 p(x)\,dx$. We now consider $q(x) = p(x) - C$. Firstly, since p is a polynomial, so is q. To see that $q \in P$:

$$\begin{aligned}\int_0^1 q(x)\,dx &= \int_0^1 \left(p(x) - C\right)\,dx \\ &= \int_0^1 p(x)\,dx - \int_0^1 C \\ &= C - C \cdot \int_0^1 1\,dx \\ &= C - C = 0.\end{aligned}$$

We also see that

$$\|f - q\|_\infty = \|f - p + C\|_\infty \leq \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Therefore, P is dense in V and so, $\overline{P} = V$.

2022, Q9

(c) Recall that $(C[0,1], \|\cdot\|_\infty)$ is complete. Therefore, it suffices to show that $(V, \|\cdot\|_\infty)$ is closed. But V is the closure of P, so it is closed.

Separation of points

Definition

Let X and Y be two sets. A set S of functions between X and Y is said to *separate points* if, for all distinct $x, y \in X$, there is a function $f \in S$ such that $f(x) \neq f(y)$.

Urysohn's Lemma

Let X be a Hausdorff and compact space. Then $C(X, \mathbb{R})$ separates points.

Separation of points

- Recall that an algebra of functions is a vector space of functions that is closed under pointwise multiplication of functions.
- A *unital* algebra is an algebra that also contains the unit constant function 1.

Stone-Weierstrass Theorem

Let X be a Hausdorff and compact space. Let $A \subseteq C(X, \mathbb{R})$ be a unital subalgebra. Then A is dense with respect to $\|\cdot\|_\infty$ if and only if A separates points.

2019, Q9

(2019, Problem 9)

Let S be the span of the set of functions $\{e^{kt}\}_{k=0,1,\ldots} \subset C[0,1]$. What are the interior, boundary, and closure of S in the uniform norm of $C[0,1]$?

Let's see if we can apply Stone-Weierstrass.

- $[0,1]$ is Hausdorff and compact.
- S is a unital subalgebra.
- Note that $f_k(t) = e^{kt}$ is injective, so $f_k(x) \neq f_k(y)$ for any pair of distinct points $x, y \in [0,1]$. Therefore, S *separates* points.

Stone–Weierstrass applies! Thus,

- The closure is $C[0,1]$.
- The interior is empty because every element of S can be arbitrarily approximated by a non-smooth but continuous function.
- The boundary is $C[0,1]$, as $\mathrm{Cl}(S) = \mathrm{Int}(S) \sqcup \mathrm{Bd}(S)$.